Regularization

DATA 201 – WEEK 9B

FALL 2026

Topics Today

- Understanding Lasso and Ridge
- Top Six Classification Algorithms

🎯 Regularization: The Goal

- **What is it?** Regularization is a technique used in regression (like **Linear Regression**) to prevent **overfitting** by adding a penalty term to the cost function.
- **Why use it?** It helps models generalize better to new, unseen data, especially when you have many predictors or when predictors are highly correlated.

Regularization

- Technique used in regression to prevent overfitting by adding a "penalty" to overly complex models
- Handles dimension reduction and multicollinearity (reduces many predictors to stabilize estimates)
- Helps models generalize better to new, unseen data, specially when there are many predictors or when predictors are highly correlated (multicollinearity)
- Mathematics of regularization
  - In linear regression, the normally the su $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ nean squared error) is minimized:
  - In regularization, a penalty term is added: $MSE + \lambda \times (penalty\ on\ coefficients)$
    - λ (lambda) is a hyperparameter that controls how strong the penalty is.
  - Two types of regularization: Ridge and Lasso

⛰️ Ridge Regression (L_2 Regularization)

- **The Penalty:** It adds a penalty based on the **square** of the magnitude of the coefficients (β). The penalty term is proportional to $\sum_{j=1}^{p} \beta_j^2$.
- **Effect on Coefficients:** It **shrinks** the coefficients toward zero.
- **Key Feature:** Ridge regression will **never set a coefficient exactly to zero**. All features remain in the model, but their influence is reduced.
- **Best For:** Situations where **all features** are relevant, but their influence needs to be controlled (e.g., when multicollinearity is present).

🔍 LASSO Regression (L_1 Regularization)

- **The Penalty:** It adds a penalty based on the **absolute value** of the magnitude of the coefficients (β). The penalty term is proportional to $\sum_{j=1}^{p} |\beta_j|$.
- **Effect on Coefficients:** It **shrinks** coefficients and can force some of them to become **exactly zero**.
- **Key Feature:** LASSO performs **automatic feature selection**. By setting coefficients to zero, it effectively removes irrelevant features from the model.
- **Best For:** Situations where you suspect only a **subset of features** are important, and you want to build a simpler, more interpretable model.

⚖️ A Shared Parameter: λ (Lambda)

- **The Control Knob:** Both techniques have a hyperparameter called λ **(lambda)**, which controls the strength of the penalty.
- $\lambda = 0$**:** The penalty term disappears, and the model is identical to standard **Linear Regression**.
- $\lambda \to \infty$**:** The penalty dominates, forcing all coefficients close to (Ridge) or exactly to (LASSO) zero, resulting in a very simple model.

1. Standard Linear Regression (OLS)

This is the baseline model that both Ridge and LASSO modify. It only minimizes the errors (residuals).

$$\text{Minimize:} \quad \text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- Where $\hat{y}_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}$

2. Ridge Regression Cost Function (L_2 Penalty)

Ridge adds a penalty term proportional to the **sum of the squared coefficients**.

$$\text{Minimize:} \quad \text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2$$

Term	Description
RSS	The error from the model (Residual Sum of Squares).
λ	**Tuning Parameter** (Lambda): Controls the strength of the penalty.
$\sum_{j=1}^{p} \beta_j^2$	The L_2 **Penalty**: The sum of the squared coefficients.

3. LASSO Regression Cost Function (L_1 Penalty)

LASSO adds a penalty term proportional to the **sum of the absolute values of the coefficients**.

$$\text{Minimize:} \quad \text{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|$$

Term	Description
RSS	The error from the model (Residual Sum of Squares).
λ	**Tuning Parameter** (Lambda): Controls the strength of the penalty.
$\sum_{j=1}^{p} \beta_j$	

- The term $\lambda \sum \beta_j^2$ (Ridge) makes the minimization surface smooth, shrinking coefficients toward zero.
- The term $\lambda \sum |\beta_j|$ (LASSO) creates "corners" on the minimization surface, which allows the optimal solution to land exactly on an axis, forcing that coefficient (β_j) to zero, thus performing feature selection.

🏔️ Ridge Regression Analogy: The Speed Bump

- **The Scenario:** Imagine you are driving a car (your **Linear Regression model**) down a road, and you tend to **overshoot** corners (overfit).
- **The Penalty (λ): Ridge Regression** is like putting a large, wide **speed bump** across the road.
- **The Effect:** This speed bump applies a **small, smooth, and continuous resistance** to *every single wheel* (coefficient).
- **The Result:** It forces you to **slow down everyone slightly** (shrink every coefficient), but **no wheel ever stops completely** (no coefficient is set to zero). All features remain in the model, but their impact is gently controlled.

🎣 LASSO Regression Analogy: The Fishing Net

- **The Scenario:** You are fishing for important predictors (features) in a large sea of possibilities.
- **The Penalty (λ): LASSO Regression** is like using a specialized **fishing net** that has a **maximum capacity**.
- **The Effect:** The net will be forced to **exclude** or **completely discard** the *smallest, lightest* or **most irrelevant fish** (features) to stay under its weight limit.
- **The Result:** Only the **most impactful** (heaviest) features are retained. The irrelevant features are completely thrown back into the water (their coefficients are set **exactly to zero**). This acts as automatic feature selection.

Other ML Algorithms Use Hyperameter Tuning

✨Linear Regression uses regularization to limit overfitting.
🎯Logistic Regression balances model complexity to improve accuracy.
🌳Decision Trees control tree depth and splits to refine precision.
🔍K-Nearest Neighbors adjusts neighbor influence for better predictions.
📈Support Vector Machines optimize decision boundaries for flexible performance.

Hyperparameter Tuning in Machine Learning

Algorithm	Hyperparameter
LINEAR REGRESSION	Alpha (λ) (for regularized parameter Ridge and Lasso)
LOGISTIC REGRESSION	C(Inverse of regularization strength), Penalty('L2' for Ridge, 'L1' for Lasso)
DECISION TREE	Criterion Max Depth Min_Samples_Split Min_Samples_Leaf
K- NEAREST NEIGHBORS	n_neighbors Weights
SUPPORT VECTOR MACHINES	C, Kernel, Gamma, Degree of tolerance

When to Use Ridge Regression?

Ridge Regression is most suitable when **all predictors are expected to contribute to the outcome and none should be excluded from the model**. It reduces overfitting by shrinking the coefficients, ensuring they don't become too large, while still keeping all the predictors in the model.

For example, when predicting house prices, features like size, number of bedrooms, location, and year built are all likely relevant. Ridge Regression ensures these features remain in the model but with reduced influence to create a balanced and robust prediction.

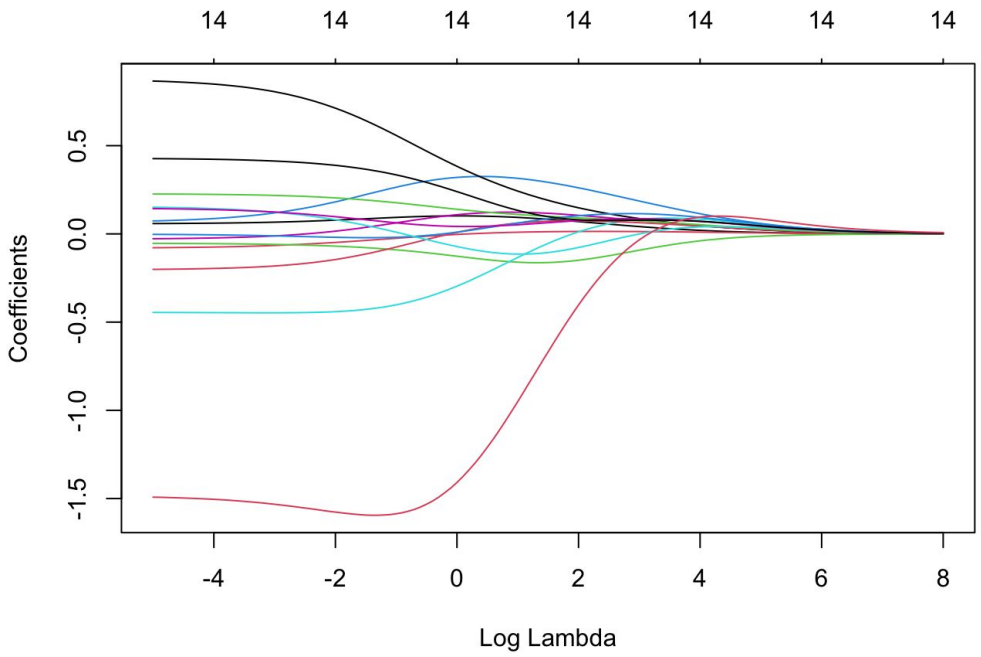
When to Use Lasso Regression?

Lasso Regression is ideal when you suspect that **only a few predictors are truly important, and the rest may add noise or redundancy**. It **performs automatic feature selection by shrinking the coefficients of less important predictors to zero, effectively removing them from the model**.

For example, in genetic research, where thousands of genes are analyzed for their effect on a disease, Lasso Regression helps by identifying only the most impactful genes and ignoring the irrelevant ones, leading to a simpler and more interpretable model.

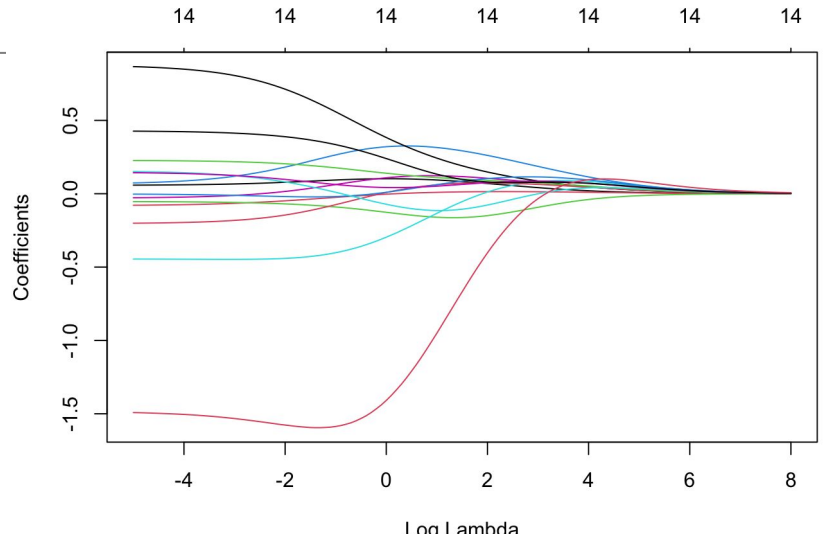
Tuning Regularization Hyperparameters (1)

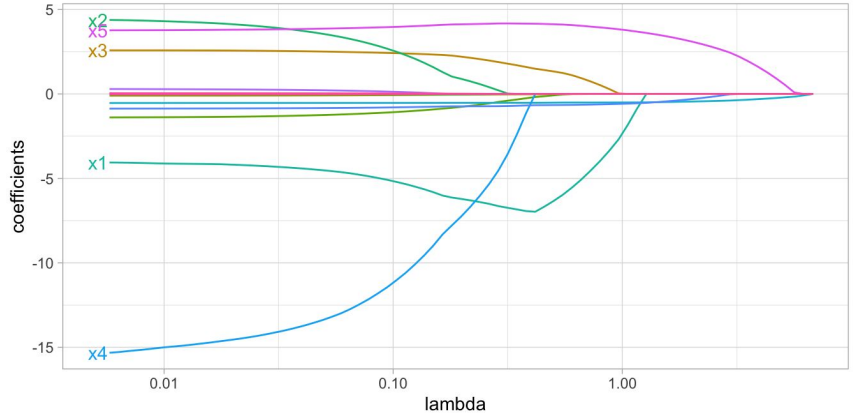
- Tuning means choosing the best hyperparameter value using data
  - λ=0 → no regularization (risk of overfitting)
  - λ=0.1
  - λ=1
  - λ=10 → too much regularization
- Use cross-validation to pick the best one
- Coefficient path plot shows how model coefficients change as λ varies
  - X – axis: regularization strength (log scale)
  - Y- axis: value of each coefficient
  - Lines: one predictor's coefficient



Tuning Regularization Hyperparameters (2)

- Ridge (L2) - top
  - Coefficients shrink smoothly to towards zero
- Lasso (L1) - bottom
  - Some coefficients become exactly zero
  - Automatic feature selection
- A coefficient path plot shows how regularization simplifies a model, and a cross-validation helps select the best level of simplification.





Coefficient Path Plot

This plot shows how the value of each feature's coefficient changes as the regularization strength (λ) is varied.

Feature	Description	Interpretation
Axes	**X-Axis:** The tuning parameter λ (often on a log scale, or scaled by the L_1/L_2 norm). **Y-Axis:** The value of the standardized coefficient (β).	As you move **left** (small λ), the coefficients approach the **unregularized OLS** values. As you move **right** (large λ), coefficients are **shrunk** toward zero.
Ridge Path	The paths for all coefficients β_j are **smooth curves** that approach zero but **never** reach it (unless λ is infinitely large).	Confirms that Ridge Regression **shrinks** all coefficients but **retains all features** in the model. All features contribute, just with a smaller impact.
LASSO Path	The paths are pieces of **straight lines** that reach the x-axis and **become exactly zero** when λ is large enough.	This confirms LASSO's key property: **automatic feature selection**. The point at which a coefficient hits zero indicates the λ value that *removes* that feature from the model.
Feature Importance	The **later** a coefficient path hits zero (moves further right on the x-axis) in the LASSO plot, the **more important** that feature is to the model.	Use this to identify the most robust and influential predictors. The variables that survive the longest are the key features.

Demo: Regularization and Coeff Paths